Menel BEREKSI

JUNIOR HYDROGEOLOGIST

Groundwater resources / Hydrogeochemistry / GIS



Gif-sur-Yvette, France | +33 6 56 66 97 89 | menel.bereksi@gmail.com

LinkedIn | Portfolio

Category B driving licence | National and international mobility

PROFILE

A Hydrology and Hydrogeology Master's graduate from Université Paris-Saclay, I combine field investigations, hydrogeochemistry and data analysis using Python and QGIS. My experience covers aquifer characterisation and water quality. I aim to contribute to groundwater resource assessment and protection, in France and internationally.

EXPERIENCE IN HYDROGEOLOGY AND GEOSCIENCES

GEOPS - Université Paris-Saclay Jan. - June 2025

Master's research internship - Environmental geochemistry

- **Water quality and contaminant transport.** Investigated arsenic and iron mobility in mine waste from Enguialès: solid-solution interactions, porosity and fluid composition.
- **Experimental protocols. 3 mine-waste samples over 6 rainfall cycles**; water-table fluctuation and hardpan leaching over **5 cycles each**; EN 12457-2, NAG and TCLP batch tests.
- **Analysis and interpretation.** pH/conductivity monitoring, ion chromatography and AAS (Fe/As); time-series processing, interpretation of SEM images and existing XRD results, and synthesis of findings.

GEOPS - Université Paris-Saclay May - June 2023

Research internship - Hydrogeology and water quality

- **Beauce multilayer aquifer.** Used **ADES and InfoTerre**, **QGIS** mapping and nitrate time series to investigate the role of low-permeability formations in groundwater protection.

SONATRACH - Research and Development Centre 2020

Master's research internship - Palynology, Boumerdès, Algeria

- Catalogued Lower Chelif palynomorphs using optical microscopy; contributed to palaeoenvironmental reconstruction.

FIELD INVESTIGATIONS - UNIVERSITY TRAINING

Université Paris-Saclay 2023-2025

- **Quantitative hydrogeology - Maisse.** Groundwater level measurements and **pumping tests in 3 boreholes**; estimation of aquifer parameters and investigation of water transfers in a multilayer system.
- **Field campaigns - Hallue and Enguialès.** Sampling, physicochemical and streamflow measurements, piezometric mapping and Porchet tests; contribution to site instrumentation at Enguialès.
- **Water quality - Crau.** Study of an aquifer affected by landfill leachate.

TECHNICAL SKILLS

Quantitative and qualitative hydrogeology

Piezometry, pumping tests, aquifer parameters, water and soil sampling.

Hydrogeochemistry and analysis

Leaching, ion chromatography, AAS, porosity, CEC; SEM/XRD data interpretation; water isotopes ($\delta^{18}O$, $\delta^{2}H$).

GIS, data and technical reporting

QGIS, Python, Excel; ADES, InfoTerre; maps, charts and synthesis of findings; Word, PowerPoint, LaTeX.

Modelling - university coursework

MODFLOW, ModelMuse, MT3DMS and PHREEQC.

Geophysics - university fieldwork

Electrical resistivity tomography (ERT), ground-penetrating radar (GPR), seismic surveying.

SCIENTIFIC COMMUNICATIONS

3 presentations in 2025: GS GCEP oral, IPSL poster and IPSL-Watercycle oral.

Co-author of an EGU 2026 conference abstract (XRF / XANES).

EDUCATION

MSc (M2) Hydrology and Hydrogeology

Université Paris-Saclay 2025

Studies 2022-2025: quantitative hydrogeology, geochemistry, modelling, geothermal energy and contaminated land.

MSc (M2) Georesources

University of Tlemcen, Algeria 2020

BSc Geology

University of Tlemcen, Algeria 2018

LANGUAGES

French C2 | English C1 | Arabic C2

OTHER EXPERIENCE AND ACTIVITIES

Naturaha - Founder

2025-2026 : Product formulation and development: ingredients, dosage, prototypes and testing; preparation for commercialisation.

PAUWES - Project assistant

Sept.-Dec. 2021 : Mapped African entrepreneurial ecosystems and coordinated scientific and educational events.

Les Amis du Malade

Co-founder, president - 2022 : Mobilised volunteer doctors and organised medical interventions.